

Detail

Y14 (2014) — English translation

A part with mass m can move without friction along vertical guides and rotate freely around a horizontal axis perpendicular to the guides and the drawing plane. The part has a through hole into which a smooth light rod is inserted, one end of which is hinged on a horizontal plane at a distance L from the base of the vertical guides (Fig. 1). Between the part and the hinge, a spring attached to them with a stiffness coefficient k is put on the rod. Free spring length $3L$. A light thread is tied to the part, which in the initial state is tensioned by a vertically upward force $F = mg$.

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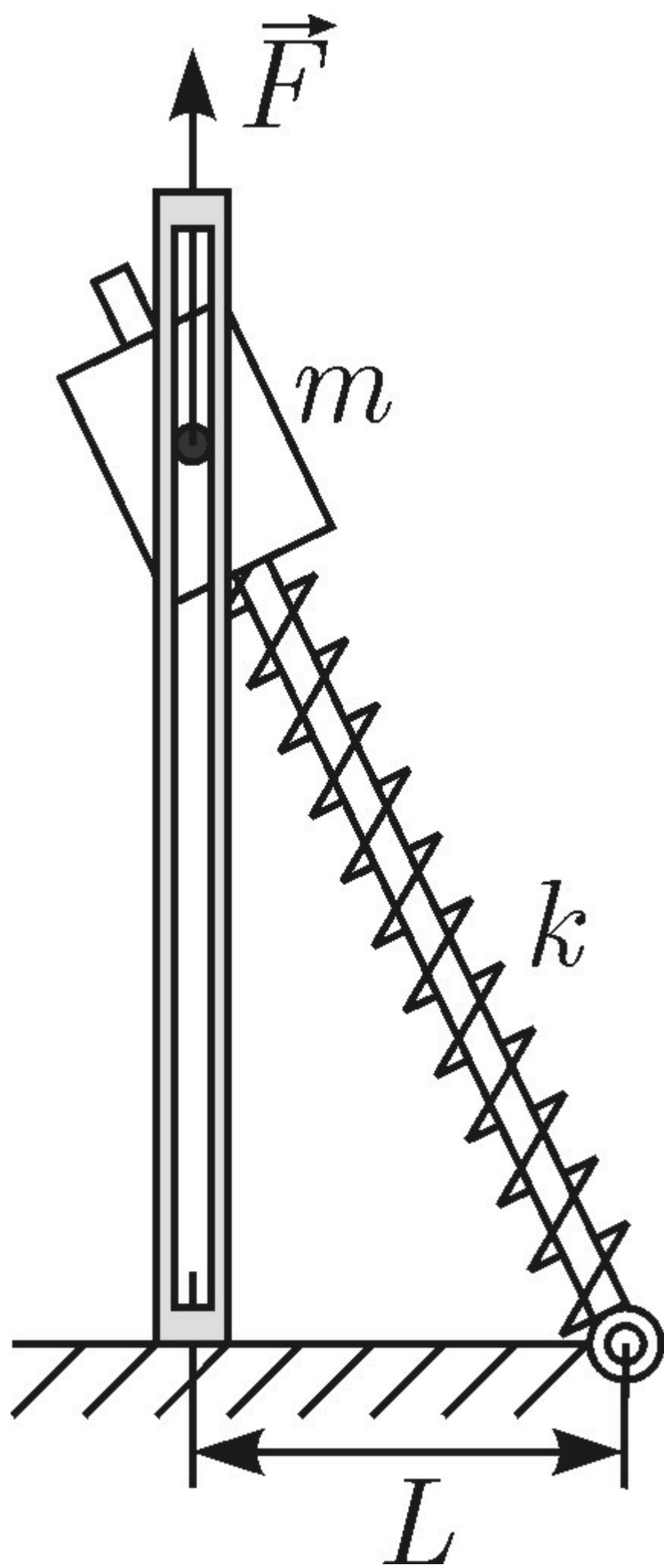


Fig. 1.

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|-----------|---|----------|
| A1 | What should be the minimum mass $m_{\min}$ of the part so that with a slow (quasi-static) decrease in force F , the part can move the rod to a horizontal position under the influence of gravity. Ignore friction. | 0 |
|-----------|---|----------|

If the force F is reduced to zero instantly, then the part can move the rod to a horizontal position with a smaller mass m_1 . The graph shows the dependence of the projection of the elastic force on the vertical axis, in relative units, on the distance from the load to the horizontal plane.

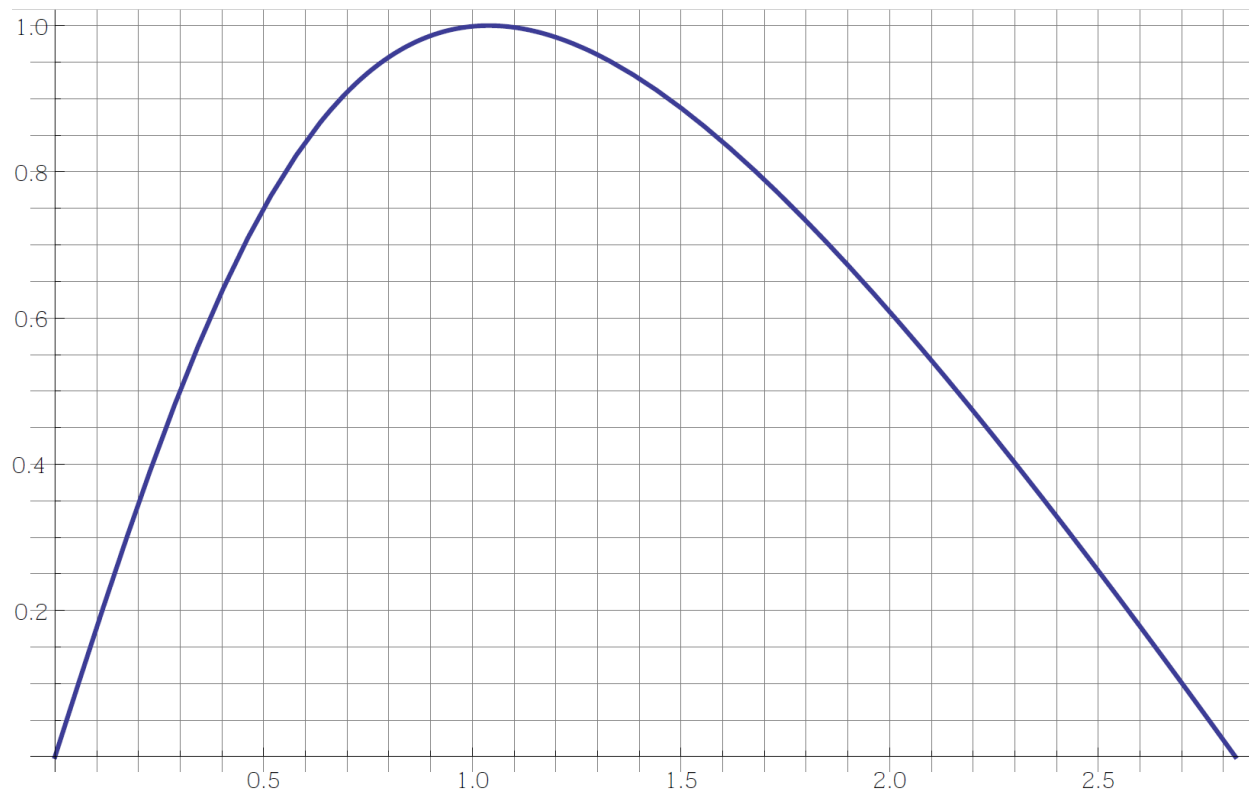


Fig. 2.

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|-----------|---|----------|
| A2 | Determine the mass of m_1 using the graph provided. | 0 |
| A3 | The force F was eliminated (the thread was cut), and the mass load m_1 was transferred to a stable equilibrium position. Determine the period of small oscillations near this position. | 0 |

Source: <https://pho.rs/p/3963>